

# Prompt Conflict Geometry

## DX Terminal Highlights Brief

April 17, 2026

### MAIN TAKEAWAY

We can now reliably read internal strategy/settings disagreement from the model across three execution dimensions: `trade_size`, `risk_preference`, and `diversification_preference`. The strongest pair is `trade_size` and `risk_preference`, but all three families share meaningful geometry. In practical terms, DX Terminal now has a credible path toward detecting when the model sees tension between strategy guidance and user settings, identifying which axis is in tension, and eventually steering or debugging those cases instead of treating them as black-box output quirks.

## Main Findings

- Three prompt-local conflict families now validate cleanly enough to use as internal diagnostics: `trade_size`, `risk_preference`, and `diversification_preference`.
- Text-only baselines stay at chance across the strong families, so the model is not solving these tasks by raw lexical shortcut.
- Single-family probes are very strong, with best balanced accuracy ranging from 0.9635 to 1.0000 on standard holdouts.
- The mixed joint prompt shows both size and risk conflicts can be read out from the same forward pass:
  - `size_conflict_present`: 0.9414 / 0.9862
  - `risk_conflict_present`: 0.9388 / 0.9871
- Same-capture geometry at L36 shows a real three-family structure rather than isolated benchmark-specific features:
  - risk vs `trade_size`: 0.6449
  - `diversification` vs risk: 0.4684
  - `diversification` vs `trade_size`: 0.4883
- The main technical caveat is calibration, not absence of signal: `diversification` carries a family-specific baseline offset that breaks raw threshold transfer even when ranking transfer remains strong.

## Probe Highlights

These numbers are best read in three buckets:

- **Behavior**: whether the model actually followed the intended setting on the benchmark rows
- **Standard probe**: how well a linear readout decodes conflict under the normal lexical holdouts
- **Strict**: the harder novelty test, where the readout has to survive a more aggressive split

The important pattern is that all three families are strongly readable, but they are not equally clean behaviorally. `trade_size` is the anchor, `risk_preference` is representationally strong but behaviorally asymmetric, and `diversification_preference` is the cleanest asset-selection family so far.

| Family / Readout           | Behavior                      | Standard Probe                   | Strict          | Note                               |
|----------------------------|-------------------------------|----------------------------------|-----------------|------------------------------------|
| trade_size                 | ~97% conflict                 | 0.9948 / 1.0000                  | 0.9896 / 1.0000 | cleanest benchmark                 |
| risk_preference            | 100% aligned                  | 0.9844 / 0.9937                  | 0.8854 / 0.9119 | behaviorally asymmetric            |
| diversification_preference | 100% aligned / 85.4% conflict | 1.0000 / 1.0000 strategy holdout | 0.8333 / 0.8819 | portfolio-conditioned third family |

Format above is balanced accuracy / AUROC.

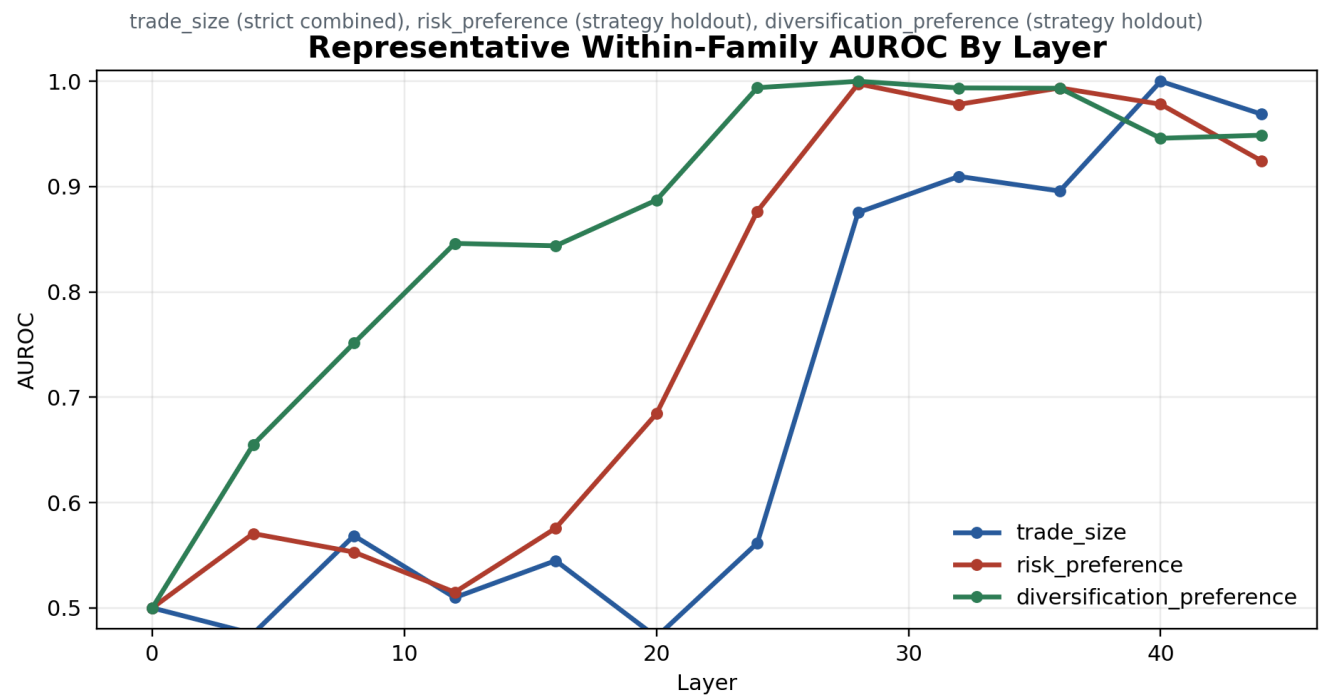


Figure 1: Representative within-family AUROC curves across layers. All three families rise sharply out of chance and become highly readable in the late middle layers rather than only at the final layer.

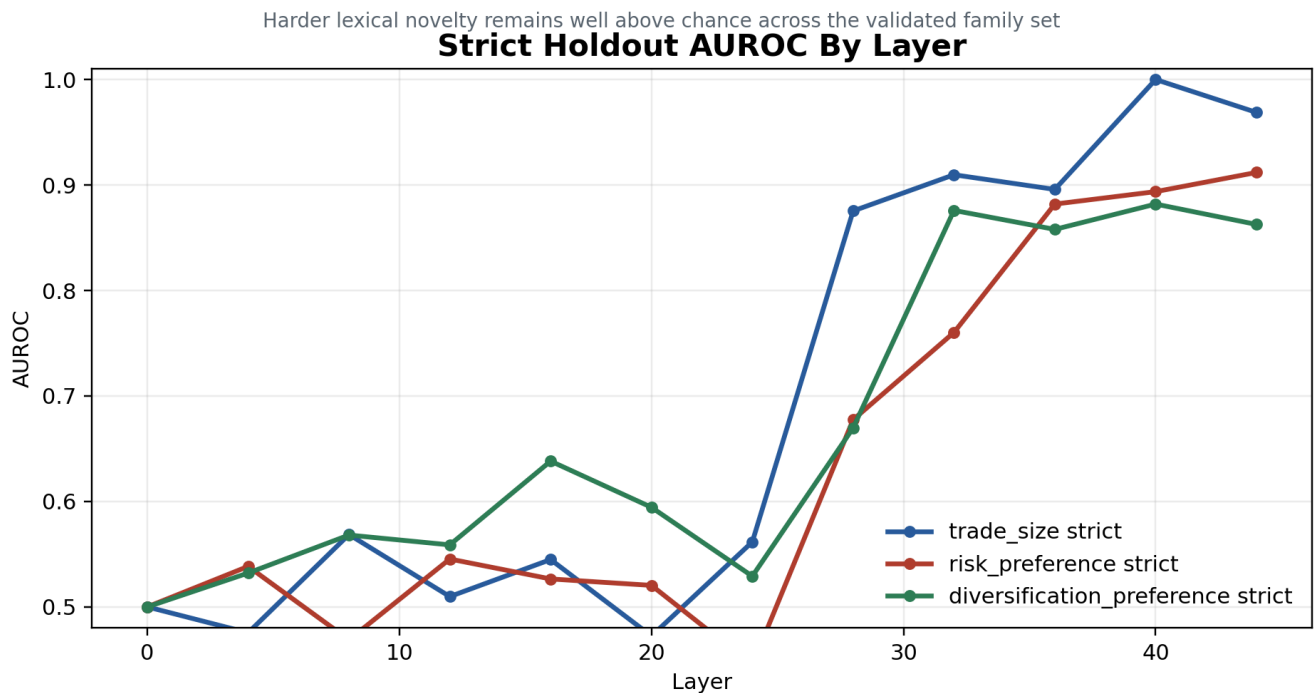


Figure 2: The harder strict split drops performance, but the validated family set still stays well above chance and peaks in the same late-middle region.

### Geometry Highlights

The geometry result is the part that makes this more useful than “we trained a few good probes.” It says the model is not building three unrelated benchmark features. It is building a family of related conflict representations with a shared internal structure.

At L36, the current best same-capture cosine picture is:

| Pair                                          | Cosine |
|-----------------------------------------------|--------|
| risk_preference vs trade_size                 | 0.6449 |
| diversification_preference vs risk_preference | 0.4684 |
| diversification_preference vs trade_size      | 0.4883 |

This is the current best high-level read:

- trade\_size and risk\_preference are the tightest pair
- diversification\_preference is not an outlier; it is a real third member
- the three families share a conflict-relevant subspace, but not identical calibration

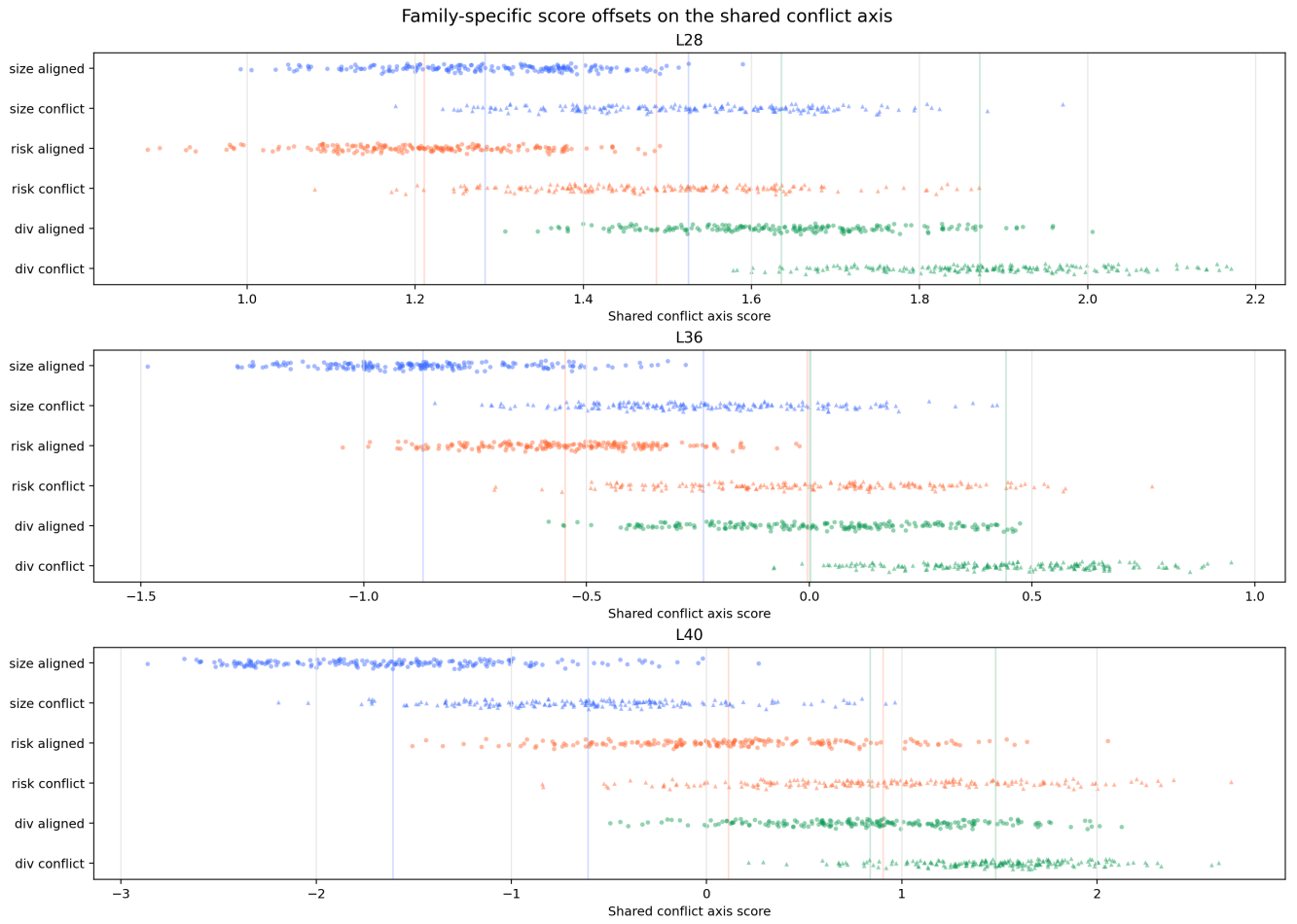


Figure 3: Shared-axis view of the three-family capture. Each family has a clear aligned-to-conflict separation, but the family baselines are shifted relative to each other. This is why cross-family AUROC can stay high while raw threshold transfer breaks.

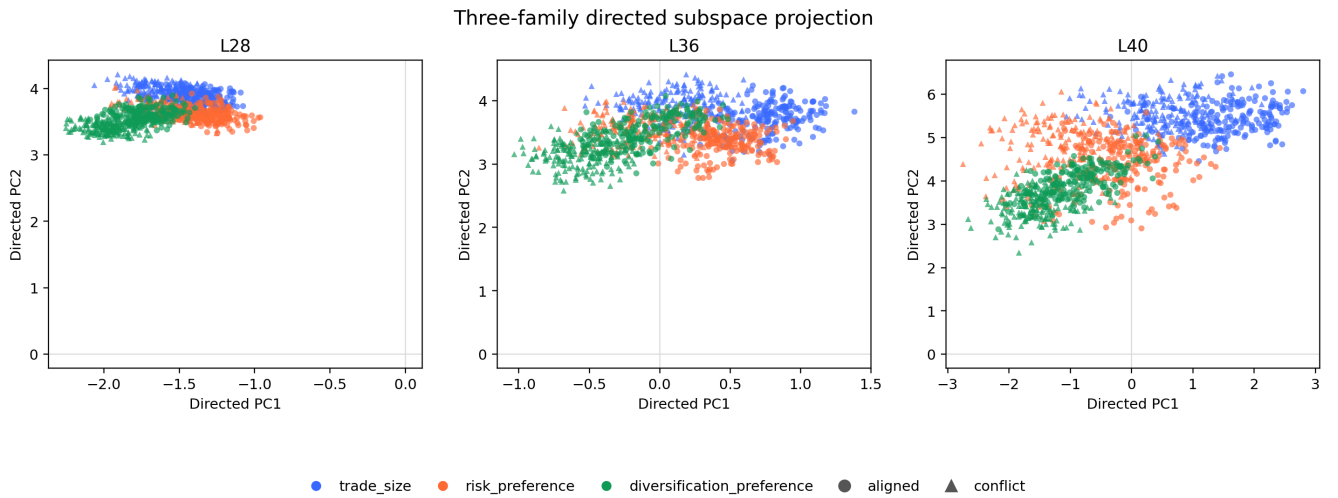


Figure 4: Directed 2D subspace view at the best geometry layer. `risk_preference` and `trade_size` remain the tightest pair, while `diversification_preference` occupies a nearby but offset region rather than an isolated one.

## Joint Prompt Result

### Simultaneous Conflict Readout

The joint-prompt run was the key compositional result. When both size and risk conflicts appear in the same prompt, both remain strongly readable from the same activation stream:

- size\_conflict\_present: 0.9414 / 0.9862
- risk\_conflict\_present: 0.9388 / 0.9871
- any\_conflict\_present: 0.9306 / 0.9503
- double\_conflict\_present: 0.8898 / 0.9687

That means the model is not just building one narrow benchmark feature per prompt type. It can maintain multiple conflict signals simultaneously.

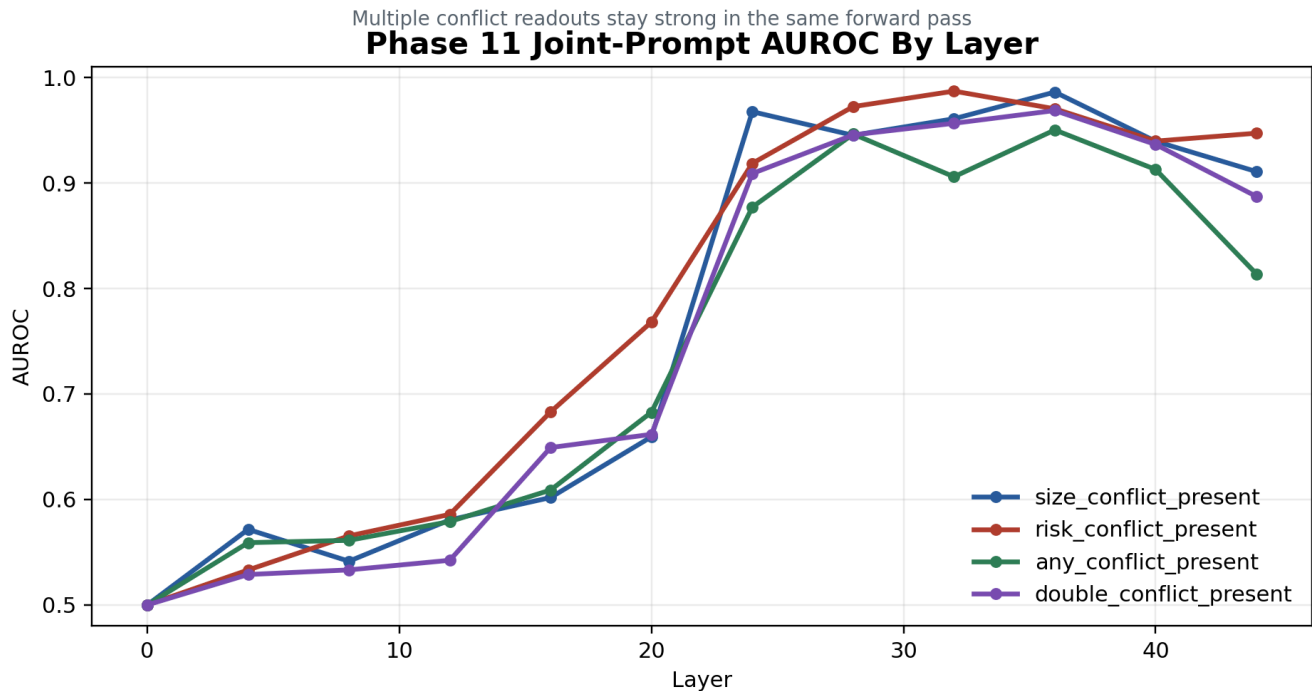


Figure 5: Joint-prompt AUROC curves. Multiple conflict labels stay strong across the same forward pass, with the cleanest readout window in the same late-middle layers as the single-family benchmarks.

### Diversification Geometry Update

The original three-family transfer workflow made diversification look farther away than it really was. The follow-up same-capture projection diagnostic showed that the bigger issue was prompt-structure offset, likely from the extra PORTFOLIO block at the last prompt token.

That changed the interpretation from:

- “diversification may be too far from the family to matter”

to:

- “diversification shares real structure, but family-specific baseline shift breaks raw threshold transfer”

That is a materially better story for DX Terminal because it means the signal is more reusable than the first transfer numbers implied.

## Why This Matters

- We can now distinguish two different failure modes:
  - the model never represented the strategy/settings disagreement
  - the model represented it, but resolved it oddly downstream
- We have axis-level readouts, not just a vague “something is off” detector. The current families already cover:
  - how much to buy
  - which asset fits the allowed risk posture
  - which asset fits the desired concentration vs broadening posture
- The geometry work gives us something stronger than a one-off classifier: a reusable conflict subspace we can score, compare, and eventually intervene on.
- This opens a path to product-facing diagnostics on real prompts:
  - detect hidden strategy/settings tension
  - identify which execution axis is in conflict
  - debug template changes mechanistically instead of by anecdote

## Recommended Next Steps

1. Validate transfer on real DX Terminal prompts, starting with the cleanest axis: `trade_size`.
2. Use the current conflict directions as monitors on real prompts before trying any intervention.
3. Return to activation patching once the infra is ready, using the shared conflict directions as the causal target.

---

Metric format is `balanced accuracy / AUROC` unless otherwise noted. All results shown here come from Qwen3-30B-A3B prompt-EOS residual analyses across Phases 09–12, with emphasis on the strongest validated findings and the corrected same-capture geometry result.